[METRIC]

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SUPERSEDING
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MILITARY SPECIFICATION

POLYVINYL ALCOHOL, GRANULAR (METRIC)

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two grades and two classes of polyvinyl alcohol.

1.2 Classification. Polyvinyl alcohol shall be of the following grades and classes, as specified (see 6.2):

Grade A - Saponification No. 125 to 167
Grade B - Saponification No. 0 to 18.7
Class 1 - For use as a dispersing agent
Class 2 - For use in munitions

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

FSC 6810

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

RR-S-366 - Sieve, Test
PPP-D-729 - Drums, Shipping and Storage, Steel, 55 Gallon (208 Liters)

MILITARY

MIL-B-117 - Bags, Sleeve and Tubing - Interior Packaging

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

D445 - Kinematic Viscosity of Transparent and Opaque Liquids (and the Calculation of Dynamic Viscosity)
D1193 - Reagent Water

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Chemical and physical characteristics. Polyvinyl alcohol shall conform to applicable chemical and physical characteristics of table I when tested as specified therein.

TABLE I. Chemical and physical characteristics

Characteristic	Grade A	Grade B	Test paragraph
Volatile matter, percent by weight, maximum	5.0	5.0	4.2.4.1
Saponification number (dry basis)	125 to 167	0 to 18.7	4.2.4.2
Ash, percent by weight, maximum (dry basis)	1.25	1.25	4.2.4.3
Absolute viscosity of 4.0 percent by weight solution at 20.0° ± 0.1°C, centipoises	35 ± 10	25 ± 7	4.2.4.4
Water insoluble matter, percent by weight, maximum (dry basis)	1.5	----	4.2.4.5
pH of 4.0 percent by weight solution	6.0 ± 1.0	6.5 ± 1.5	4.2.4.6

3.2 Particle size. Polyvinyl alcohol shall conform to the particle size characteristics of table II when tested as specified in the applicable portion of 4.2.4.7.

TABLE II. Particle size characteristics

Minimum percent by weight passing	Class 1	Class 2
No. 10 sieve	99.0	----
No. 20 sieve	95.0	----
No. 80 sieve	----	99.0

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

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4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the polyvinyl alcohol of one grade and class produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For polyvinyl alcohol test. Sampling shall be conducted in accordance with table III. A representative specimen of approximately 400 grams (g) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

TABLE III. Sampling for polyvinyl alcohol test

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.2.3 For container leakage test. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one shipping container ready for shipment. Sample shipping containers shall be examined for the following defects grouped collectively using an AQL of 4.0 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Marking incorrect, missing, or illegible
(f) Liner bag missing or not as specified (when required)
(g) Unitization not as specified

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4.2.3.2 For polyvinyl alcohol test. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.3.3 For container leakage test. The sample unit shall be one container. The sample containers selected in 4.2.2.3 shall be tested as specified in 4.2.5 using an AQL of 2.5 percent defective.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Volatile matter. Weigh, to the nearest 0.01 g, approximately 10 g of the specimen into a tared crystallization dish approximately 50 millimeters (mm) in diameter and 35 mm high. Heat in an oven at 105° to 110°C for 3 hours. Cool to room temperature in a desiccator and weigh. Calculate the percent volatile matter as follows:

$$\text{Percent volatile matter} = \frac{100\ (A - B)}{(A - C)}$$

where: A = Weight of dish and specimen before heating in grams,
B = Weight of dish and specimen after heating in grams, and
C = Weight of empty dish in grams.

4.2.4.2 Saponification number. Weigh to the nearest milligram (mg) approximately 1 g of specimen previously dried at 100° ± 5°C for 3 hours, and quantitatively transfer to a 250-milliliter (ml), wide-mouth Erlenmeyer flask having a standard taper ground glass connection. Add 50 ml of methanol and swirl the flask. Add 50 ml of water and swirl again. Add 1 ml of a 0.1 percent ethanol solution of phenolphthalein and neutralize with 0.1N sodium hydroxide solution or 0.1N hydrochloric acid, as required. Add 10.0 ml of 0.5N sodium hydroxide solution and a few glass beads. Connect the flask to a reflux condenser. Prepare a second flask for use as a blank as specified for the first flask except omit the specimen. Reflux the specimen and the blank for 1 hour. Allow the flasks to cool to room temperature and titrate each with 0.1N hydrochloric acid using phenolphthalein as indicator. Calculate the saponification number (S) as follows:

$$S = \frac{5{,}610\ A\ (B - C)}{W}$$

where: S = Number of milligrams of potassium hydroxide required to saponify 1 g of dried specimen,
A = Normality of hydrochloric acid used,
B = Milliliters of hydrochloric acid used to titrate blank,
C = Milliliters of hydrochloric acid used to titrate specimen, and
W = Weight of dried specimen in grams.

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4.2.4.3 Ash. Weigh to the nearest milligram approximately 2 g of the specimen into a tared platinum crucible. Ignite over a burner at low heat until most of the carbon is burned off. Cool to room temperature, add 1 ml of 1 to 1 sulfuric acid, carefully evaporate to dryness over a burner, and then place in a muffle furnace at 800° ± 20°C for 30 minutes. Cool to room temperature in a desiccator and weigh. Calculate the percent ash (as sodium oxide) as follows:

$$\text{Percent ash} = \frac{4{,}360\ A}{W\ (100 - B)}$$

where: A = Weight of ash in grams,
B = Percent volatile matter (see 4.2.4.1), and
W = Weight of specimen in grams.

4.2.4.4 Absolute viscosity.

(a) Preparation of solution. Add 300 ml of water to a tared 500-ml wide-mouth Erlenmeyer flask equipped with a stirrer. Slowly add 15.0 ± 0.1 g (on a dry basis) of the specimen to the water in the flask with sufficient agitation to prevent the specimen from forming an agglomerated mass. Heat the solution to 90°C with continued gentle agitation for at least 1 hour until the specimen is completely dissolved. Cool the solution and determine the solids content by measuring duplicate amounts of about 10 ml each into separate tared aluminum foil or moisture dishes (approximately 58 mm in diameter and 18 mm in depth) that were previously dried by heating and cooled in a desiccator. After obtaining the solution weight to the nearest 0.01 g, dry in an oven at 105° to 110°C to constant weight, cooling to room temperature in a desiccator prior to each weighing. Calculate the percent solids as follows:

$$\text{Percent solids} = \frac{100\ W}{P}$$

where: W = Weight of solids after drying in grams, and
P = Weight of solution in grams.

Prepare a 4.0-percent solution by appropriate dilution of 200 g of the solution prepared above.

(b) Procedure. Determine the viscosity of the 4.0-percent solution prepared in (a) at 20°C in accordance with ASTM D445, by employing a No. 300 Ostwald type Cannon-Fenske viscometer that has a calibrated instrument constant. Fill the viscometer by transferring 10 ml of solution from a 10-ml pipet that was filled by drawing up the solution through a clean piece of cotton held over the bottom tip. Mount the viscometer in a thermostated bath at 20.0° ± 0.1°C, and measure the flow time through the viscometer capillary at least twice to obtain agreement of results. Determine the density of the solution at the same temperature as the viscosity to the nearest 0.001 g per cubic centimeter with a hydrometer or other applicable method. (Density values in grams per milliliter are numerically equal.)

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(c) Calculation. Calculate the kinematic viscosity in centistokes from the measured flow time as specified in ASTM D445. Calculate the absolute (dynamic) viscosity in centipoises from the kinematic viscosity as specified in ASTM D445.

4.2.4.5 Water insoluble matter. Adjust the pH of 50 g of the 4.0-percent solution prepared in 4.2.4.4(a) so that is is just alkaline to phenolphthalein. Dilute with 450 ml of water. Filter the solution through a tared No. 100 mesh screen, and wash the residue after filtration with three separate 50-ml portions of water. Dry the screen in an oven at 105° to 110°C to constant weight, cooling in a desiccator to room temperature prior to each weighing. Calculate the percent water insoluble matter as follows:

$$\text{Percent water insoluble matter} = 50 (A - B)$$

where: A = Weight of screen and residue after drying in grams, and
B = Weight of screen in grams.

4.2.4.6 pH of 4.0-percent solution. Determine the pH, to the nearest tenth, of the 4.0-percent solution prepared in 4.2.4.4(a), using a suitable pH meter and electrode compensated for temperature change.

4.2.4.7 Particle size.

(a) Class 1. Use sieves conforming to RR-S-366. Nest a No. 10 sieve on top of a No. 20 sieve and place a receiving pan on the bottom. Place 100.0 g of the specimen on the top sieve, cover, and place in a mechanical shaker geared to produce 300 ± 20 gyrations and 150 ± 10 taps of the striker per minute. Shake for 15 minutes, weigh the specimen retained on each sieve and on the bottom pan and calculate the percent specimen passing through each sieve.

(b) Class 2. Determine percent passing as specified in (a) except use a No. 80 sieve conforming to RR-S-366 instead of the No. 10 and No. 20 sieves.

4.2.5 Container leakage test. Place the sample drum on its side and roll it a distance of approximately 2 meters forwards and 2 meters backwards. Open the drum and examine the bag liner for evidence of leakage (sifting of contents from the bag).

5. PACKAGING

5.1 Unit packing. Unit packing shall be level A, B, or commercial as specified (see 6.2).

5.1.1 Level A. A quantity of 45.4 (+ 0.5 or -0) kilograms of polyvinyl alcohol shall be unit packed in a removable head steel drum furnished with a bag liner. The drum shall conform to Uniform Freight Classification (UFC) Rule

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40, and its exterior shall be coated as specified for the steel drums of PPP-D-729. The bag liner shall be designed to fit the drum and shall conform to MIL-B-117, type I, class E, style optional. The bag shall be sealed as specified in MIL-B-117, and the drum shall be closed in accordance with the drum specification. There shall be no evidence of sifting of contents from the bag when the filled drum is tested as specified in 4.2.5.

5.1.2 Level B. Polyvinyl alcohol shall be unit packed as specified in 5.1.1 except that the drum shall be a fiber drum conforming to UFC Rule 51.

5.1.3 Commercial. Polyvinyl alcohol shall be unit packed commercially in accordance with MIL-STD-1188.

5.2 Unitization. Level A and B packs of polyvinyl alcohol shall be palletized in accordance with MIL-STD-147. Commercial packs shall be palletized in a manner to assure carrier acceptance and safe delivery at destination.

5.3 Marking. Level A and B packs, unit packs, and pallet loads shall be marked in accordance with MIL-STD-129. Commercial packs and unit packs shall be marked in accordance with MIL-STD-1188. Commercial pallet loads shall be marked to assure proper identification and prompt delivery at destination.

6. NOTES

6.1 Intended use. Grade A, class 1 polyvinyl alcohol is intended for use as a dispersing agent in clothing impregnation solution formulations. Grade B, class 2 polyvinyl alcohol is intended for use in a burning mix formulation as a delay element for bombs.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Grade and class of polyvinyl alcohol required (see 1.2)
(c) Level of unit packing required (see 5.1)

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

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